

Introduction to Probability and Statistics

Exercises Lecture 6

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Exercise 1: Adapted from Chapter 8, problem 25. It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of the current gain is 200 and **the standard deviation is unknown, but estimated from the sample standard deviation** to be 35, would the claim be rejected at the 5 percent level of significance if

- a) The sample size is 25?
- b) The sample size is 64?

Note that the standard deviation is now unknown. How does this change the problem when compared to the one from lecture 4?

Exercise 2: Chapter 8, problem 43

A question of medical importance is whether jogging leads to a reduction in one's heart rate. To test this hypothesis, 8 nonjogging volunteers agreed to begin a 1-month jogging program. After the month, their pulse rates were determined and compared with their earlier values. If the data are as follows, can we conclude that jogging has an effect on the pulse rates with a 5 percent level of significance?

Subject	1	2	3	4	5	6	7	8
Pulse rate before	74	86	98	102	78	84	79	70
Pulse rate after	70	85	90	110	71	80	69	74

Several of the exercises were taken from the book "Introduction to Probability and Statistics for Engineers and Scientists" by Sheldon M. Ross, 3rd Ed.

Tips: Which type of test should you use? The paired t-test. How should you formulate the null-hypothesis so that it represents that jogging has no effect on pulse rates?

Exercise 3:

Two different types of cable insulation have been tested. We need to determine the voltage level at which failures occur. The tests revealed that the individual cables failed at the following voltage levels.

TABLE I
VOLTAGE LEVEL AT WHICH FAILURES OCCURRED

Type A cables		Type B cables	
36	54	52	60
44	52	64	44
41	37	38	48
53	51	68	46
38	44	66	70
36	35	52	62
34	44		

We know that the voltage level that cables of type A can withstand is normally distributed with unknown mean μ_A and known variance $\sigma_A^2 = 40$. We also know that the voltage level that cables of type B can withstand is normally distributed with unknown mean μ_B and known variance $\sigma_B^2 = 100$.

With these data, can you reject the hypothesis $\mu_A = \mu_B$ with

- a) 5 percent level of significance?
- b) 10 percent level of significance?
- c) 1 percent level of significance?

Tips: Which type of test should you use? What happens to the result of a test when we increase or decrease the level of significance?

Exercise 4: Chapter 9, problem 12

The following data set presents the heights of 12 male law school classmates whose law school examination scores were roughly equal. It also gives their annual salaries 5 years after graduation. Each of them went into corporate law. The height is in inches and the salary in units of \$1,000.

Height	Salary
64	91
65	94
66	88
67	103
69	77
70	96
72	105
72	88
74	122
74	102
75	90
76	114

- a) Do the above data establish the hypothesis that a lawyer's salary is related to his height?
Use the 5 percent level of significance.
- b) What was the null hypothesis in part a)